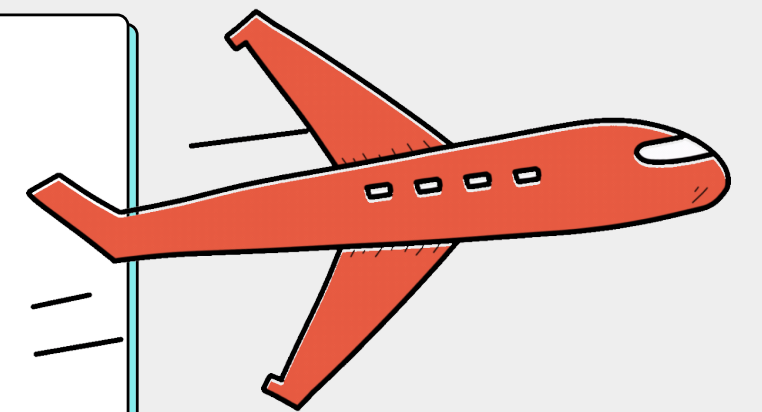


[Application] - Database Systems

Supervised by:
Eng. Ibrahim Abdallah

Q Airport Management System



Lockdown Crew



Q Creation of ERD

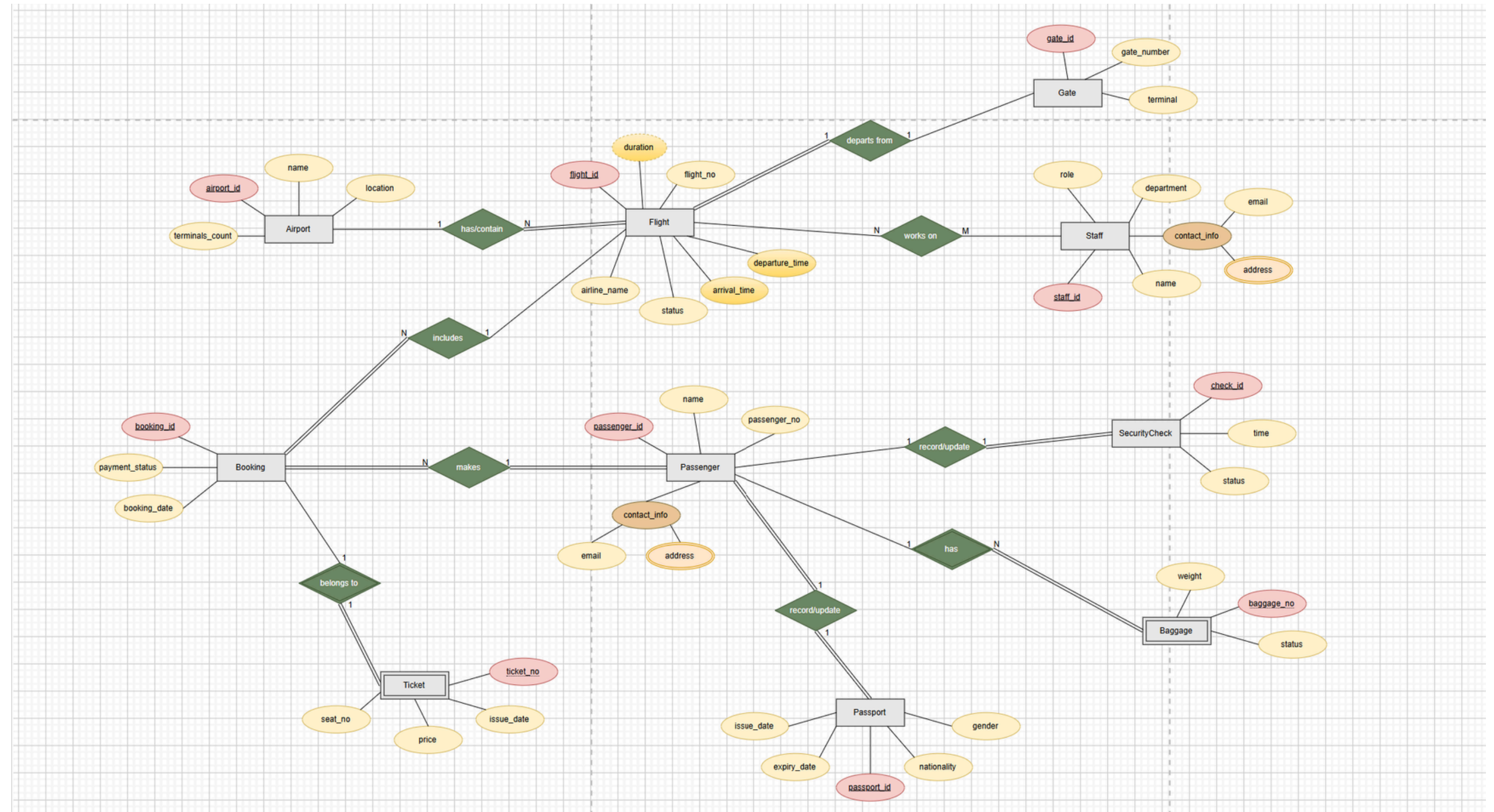


Entity Relationship Diagram



DATABASE

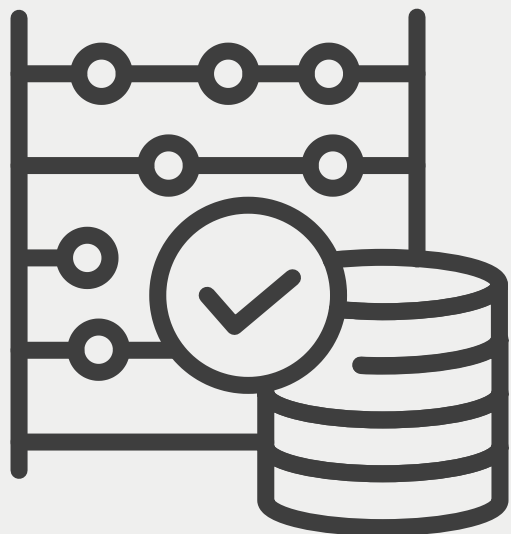
ERD

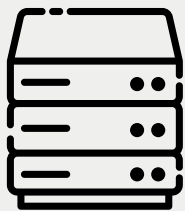


Lockdown Crew

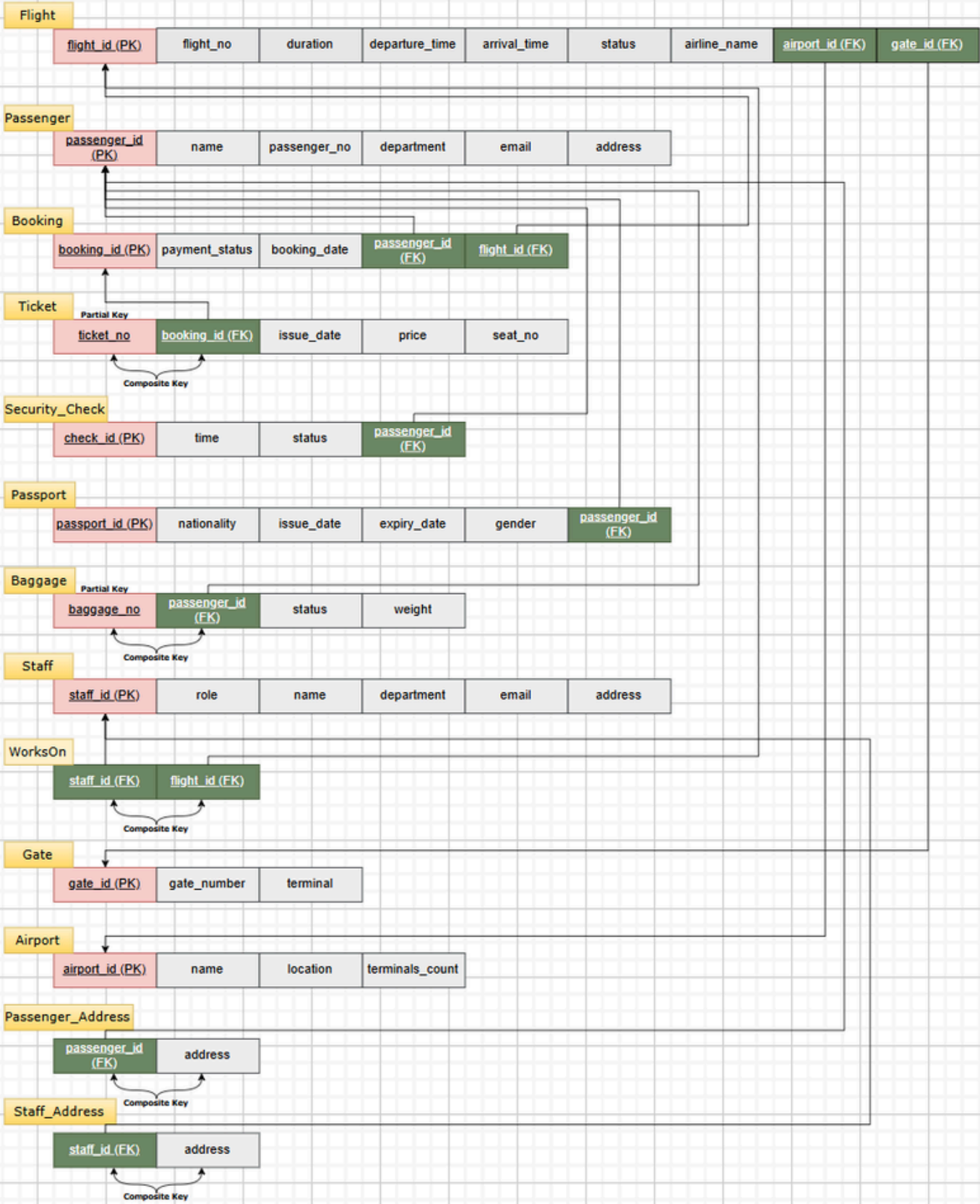


Q Creation of Schema





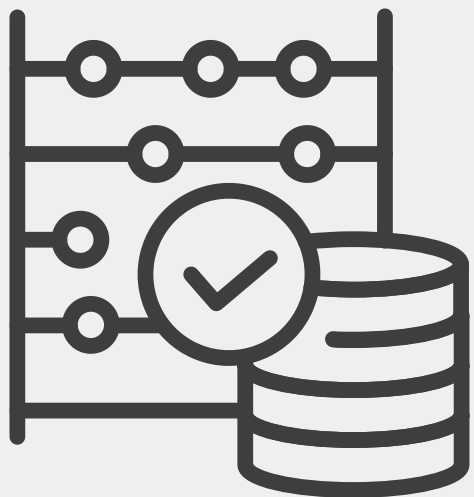
SCHEMA





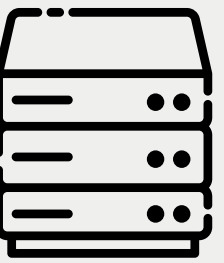
- *presented by Mai & Mohamed.*

Q Create Tables





Airport & Gate



Airport

```
USE FINAL_AIRPORT

create table airport (
    airportId int primary key,
    name varchar(100) not null,
    location varchar(100) not null,
    terminalsCount int not null
);

insert into airport values
(1, 'Cairo Airport', 'Cairo', 3),
(2, 'Alex Airport', 'Alexandria', 2),
(3, 'Aswan Airport', 'Aswan', 1),
(4, 'Luxor Airport', 'Luxor', 2),
(5, 'Hurghada Airport', 'Hurghada', 3);
```



Gate

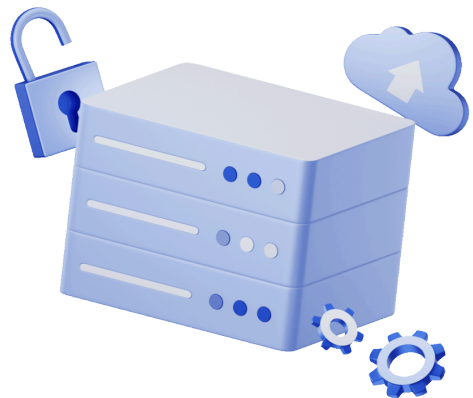
```
create table gate (
    gateId int primary key,
    gateNumber varchar(20) not null,
    terminal varchar(20) not null
);

insert into gate values
(1, 'G1', 'T1'),
(2, 'G2', 'T1'),
(3, 'G3', 'T2'),
(4, 'G4', 'T2'),
(5, 'G5', 'T3');
```



Flight

```
create table flight (  
    flightId int primary key,  
    flightNo varchar(50) not null,  
    duration varchar(50),  
    departureTime datetime not null,  
    arrivalTime datetime not null,  
    status varchar(50),  
    airlineName varchar(100) not null,  
    airportId int not null,  
    gateId int not null,  
    foreign key (airportId) references airport(airportId)  
        on delete cascade  
        on update cascade,  
    foreign key (gateId) references gate(gateId)  
        on delete no action  
        on update cascade  
);  
  
insert into flight values  
(1, 'MS101', '2h', '2025-01-01 08:00', '2025-01-01 10:00', 'on time', 'EgyptAir', 1, 1),  
(2, 'MS102', '1.5h', '2025-01-02 09:00', '2025-01-02 10:30', 'delayed', 'EgyptAir', 2, 2),  
(3, 'QR201', '3h', '2025-01-03 11:00', '2025-01-03 14:00', 'on time', 'Qatar Airways', 3, 3),  
(4, 'EK301', '4h', '2025-01-04 13:00', '2025-01-04 17:00', 'cancelled', 'Emirates', 4, 4),  
(5, 'SV401', '2.5h', '2025-01-05 15:00', '2025-01-05 17:30', 'on time', 'Saudi Airlines', 5, 5);
```





Passenger



Passenger

```
create table passenger (  
    passengerId int primary key,  
    name varchar(100) not null,  
    passengerNo varchar(50) not null,  
    department varchar(50),  
    email varchar(100),  
    address varchar(200) not null  
);  
  
insert into passenger values  
(1, 'Ahmed Ali', 'P1001', 'IT', 'ahmed@gmail.com', 'Cairo'),  
(2, 'Mona Hassan', 'P1002', 'HR', 'mona@gmail.com', 'Giza'),  
(3, 'Omar Salah', 'P1003', 'Finance', 'omar@gmail.com', 'Alex'),  
(4, 'Sara Adel', 'P1004', 'Sales', 'sara@gmail.com', 'Aswan'),  
(5, 'Youssef Kamal', 'P1005', 'Marketing', 'youssef@gmail.com', 'Luxor');
```



Q Booking & Ticket

Booking

```
create table booking (  
    bookingId int primary key,  
    paymentStatus varchar(50) not null,  
    bookingDate date not null,  
    passengerId int not null,  
    flightId int not null,  
    foreign key (passengerId) references passenger(passengerId)  
        on delete cascade  
        on update cascade,  
    foreign key (flightId) references flight(flightId)  
        on delete cascade  
        on update cascade  
);  
  
insert into booking values  
(1, 'paid', '2024-12-01', 1, 1),  
(2, 'paid', '2024-12-02', 2, 2),  
(3, 'pending', '2024-12-03', 3, 3),  
(4, 'paid', '2024-12-04', 4, 4),  
(5, 'cancelled', '2024-12-05', 5, 5);
```

Ticket

```
create table ticket (  
    ticketNo int,  
    bookingId int not null,  
    issueDate date not null,  
    price decimal(10,2) not null,  
    seatNo varchar(10) not null,  
    primary key (ticketNo, bookingId),  
    foreign key (bookingId) references booking(bookingId)  
        on delete cascade  
        on update cascade  
);  
  
insert into ticket values  
(101, 1, '2024-12-01', 1500, 'A1'),  
(102, 2, '2024-12-02', 1200, 'B2'),  
(103, 3, '2024-12-03', 1800, 'C3'),  
(104, 4, '2024-12-04', 2000, 'D4'),  
(105, 5, '2024-12-05', 1600, 'E5');
```

Q SecurityCheck & Passport



Security Check

```
= create table securityCheck (  
    checkid int primary key,  
    time datetime not null,  
    status varchar(50) not null,  
    passengerId int not null,  
    foreign key (passengerId) references passenger(passengerId)  
        on delete cascade  
        on update cascade  
);  
  
= insert into securityCheck values  
(1, '2025-01-01 07:30', 'clear', 1),  
(2, '2025-01-02 08:30', 'clear', 2),  
(3, '2025-01-03 10:30', 'hold', 3),  
(4, '2025-01-04 12:30', 'clear', 4),  
(5, '2025-01-05 14:30', 'clear', 5);
```

Passport

```
create table passport (  
    passportId int primary key,  
    nationality varchar(50) not null,  
    issueDate date not null,  
    expiry_date date not null,  
    gender varchar(10) not null,  
    passengerId int not null,  
    foreign key (passengerId) references passenger(passengerId)  
        on delete cascade  
        on update cascade  
);  
  
insert into passport values  
(1, 'Egyptian', '2020-01-01', '2030-01-01', 'male', 1),  
(2, 'Egyptian', '2019-02-01', '2029-02-01', 'female', 2),  
(3, 'Egyptian', '2018-03-01', '2028-03-01', 'male', 3),  
(4, 'Egyptian', '2021-04-01', '2031-04-01', 'female', 4),  
(5, 'Egyptian', '2022-05-01', '2032-05-01', 'male', 5);
```

Q Baggage & Staff

Baggage

```
create table baggage (  
    bag_no int,  
    passengerId int not null,  
    status varchar(50),  
    weight int not null,  
    primary key (bag_no, passengerId),  
    foreign key (passengerId) references passenger(passengerId)  
        on delete cascade  
        on update cascade  
);  
  
insert into baggage values  
(1,1,'checked',20),  
(2,2,'checked',18),  
(3,3,'lost',22),  
(4,4,'checked',25),  
(5,5,'checked',19);
```

Staff

```
create table staff (  
    staffId int primary key,  
    role varchar(50) not null,  
    name varchar(100) not null,  
    department varchar(50),  
    email varchar(100),  
    address varchar(200)  
);  
  
insert into staff values  
(1,'Pilot','Ali Mahmoud','Flight','ali@gmail.com','Cairo'),  
(2,'Co-Pilot','Hany Mostafa','Flight','hany@gmail.com','Giza'),  
(3,'Engineer','Karim Said','Maintenance','karim@gmail.com','Alex'),  
(4,'Security','Mohamed Adel','Security','mohamed@gmail.com','Aswan'),  
(5,'Hostess','Nour Hassan','Cabin','nour@gmail.com','Luxor');
```

Q Works On & Passenger Address

Works On

```
create table works_on (  
    staffId int not null,  
    flightId int not null,  
    primary key (staffId, flightId),  
    foreign key (staffId) references staff(staffId)  
        on delete cascade  
        on update cascade,  
    foreign key (flightId) references flight(flightId)  
        on delete cascade  
        on update cascade  
);  
  
insert into works_on values  
(1,1),  
(2,2),  
(3,3),  
(4,4),  
(5,5);
```

Passenger Address

```
create table passenger_address (  
    passengerId int not null,  
    address varchar(200) not null,  
    primary key (passengerId, address),  
    foreign key (passengerId) references passenger(passengerId)  
        on delete cascade  
        on update cascade  
);  
  
insert into passenger_address values  
(1, 'Cairo'),  
(2, 'Giza'),  
(3, 'Alexandria'),  
(4, 'Aswan'),  
(5, 'Luxor');
```



Staff Address

Staff Address

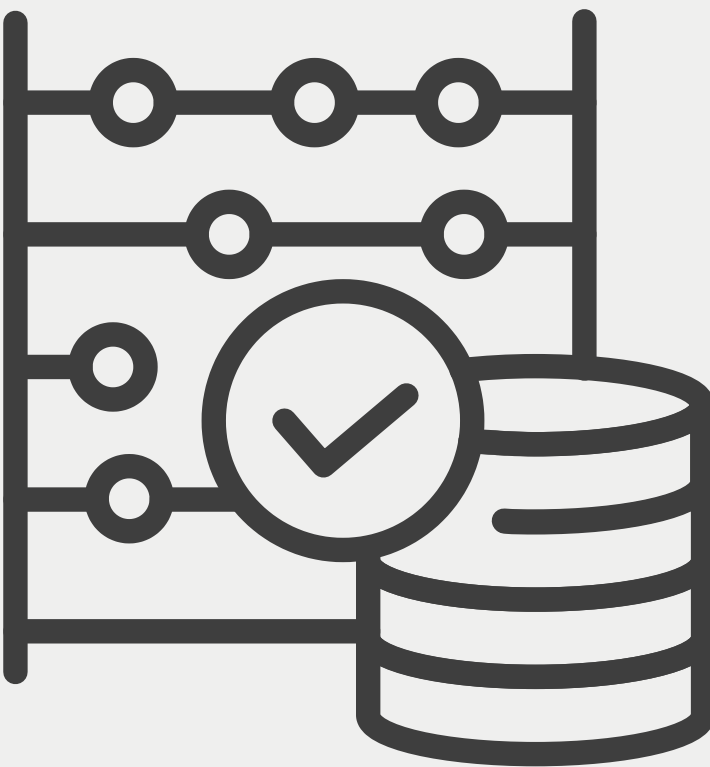
```
= create table staff_address (  
    staffId int not null,  
    address varchar(200) not null,  
    primary key (staffId, address),  
    foreign key (staffId) references staff(staffId)  
        on delete cascade  
        on update cascade  
);  
  
= insert into staff_address values  
    (1, 'Cairo'),  
    (2, 'Giza'),  
    (3, 'Alexandria'),  
    (4, 'Aswan'),  
    (5, 'Luxor');
```

Lockdown Crew



- Presented by Shahd & Yousef

Q QUERY



Q AIRPORT TABLE QUERIES



```
-- AIRPORT TABLE QUERIES  
-----  
select * from airport;  
select name, location from airport;  
select * from airport where terminalsCount > 2;  
select * from airport order by terminalsCount desc  
select * from airport where location = 'Cairo';  
select name as airport_name, location as city from airport;  
select count(*) as total_airports from airport;  
select max(terminalsCount) as max_terminals from airport;  
select * from airport  
where terminalsCount > (select avg(terminalsCount) from airport);
```

Q GATE TABLE QUERIES



```
-- GATE TABLE QUERIES

select * from gate;
select * from gate where terminal = 'T1';
select terminal, count(*) as total_gates
from gate group by terminal;
select gateNumber + '-' + terminal as gate_code from gate;
```

Q FLIGHT TABLE QUERIES



```
-- FLIGHT TABLE QUERIES

select * from flight;
select * from flight where status = 'on time';
select flightNo, airlineName from flight;
select * from flight
where airportId in (select airportId from airport where terminalsCount > 2);

select flight.flightNo, airport.name as airport_name, flight.status
from flight inner join airport on flight.airportId = airport.airportId;

select flight.flightNo, gate.gateNumber
from flight inner join gate on flight.gateId = gate.gateId;
```

Q PASSENGER TABLE QUERIES



```
-- PASSENGER QUERIES
```

```
select * from passenger;
```

```
select * from passenger where address in ('Cairo', 'Giza');
```

```
select * from passenger where department = 'IT';
```

```
select address, count(*) as total_passengers  
from passenger group by address;
```

```
select *  
from passenger  
where passengerId in (select passengerId from booking where paymentStatus = 'paid');
```



Q BOOKING TABLE QUERIES



```
-- BOOKING QUERIES

select * from booking;
select * from booking where paymentStatus = 'paid';

select paymentStatus, count(*) as count_status
from booking group by paymentStatus;

select booking.bookingId, passenger.name, flight.flightNo, booking.paymentStatus
from booking
inner join passenger on booking.passengerId = passenger.passengerId
inner join flight on booking.flightId = flight.flightId;
```

Q TICKET & SECURITY CHECK TABLE QUERIES



```
-- TICKET QUERIES
-----

select * from ticket;
select * from ticket where price > 1500;
select bookingId, count(*) as total_tickets
from ticket group by bookingId;

-----

-- SECURITY CHECK QUERIES
-----

select * from securityCheck;
select * from securityCheck where status = 'hold';
select passengerId, count(*) as checks
from securityCheck group by passengerId;
```

Q PASSPORT & BAGGAGE TABLE QUERIES



```
-----  
-- PASSPORT QUERIES  
-----  
select * from passport;  
select * from passport where gender = 'male';  
select nationality, count(*) from passport group by nationality;  
  
select *  
from passport  
where expiry_date < '2030-01-01';  
  
-----  
-- BAGGAGE QUERIES  
-----  
select * from baggage;  
select * from baggage where status = 'lost';  
select passengerId, sum(weight) as total_weight  
from baggage group by passengerId;
```

Q STAFF & WORKS_ON TABLE QUERIES



```
-- STAFF QUERIES
-----

select * from staff;
select * from staff where role = 'Pilot';
select department, count(*) as total_staff
from staff group by department;

-- works_on QUERIES
-----

select * from works_on;

select staff.name, flight.flightNo
from works_on
inner join staff on works_on.staffId = staff.staffId
inner join flight on works_on.flightId = flight.flightId;
```

Q PASSENGER & STAFF ADDRESS TABLE QUERIES



```
-- PASSENGER ADDRESS QUERIES

select * from passenger_address;

select address, count(*) as occurrences
from passenger_address
group by address;

-- STAFF ADDRESS QUERIES

select * from staff_address;

select address, count(*) as occurrences
from staff_address group by address;
```

Lockdown Crew



- Presented by Zynab



VIEWS



Q FLIGHT VIEWS



```
-- FLIGHT VIEWS
```

```
create view flight_info as  
select
```

```
    flightNo, duration, departureTime, arrivalTime, status,  
    airlineName, gateNumber, airport.name as airport_name
```

```
from flight
```

```
inner join gate on flight.gateId = gate.gateId
```

```
inner join airport on flight.airportId = airport.airportId;
```

```
create view upcoming_flights as  
select * from flight
```

```
where departureTime > '2024-12-31';
```

```
create view ontime_flights as
```

```
select * from flight where status = 'on time';
```

Q PASSENGER & BOOKING VIEWS

```
-- PASSENGER VIEWS

-- PASSENGER VIEWS

create view passenger_info as
select passengerId, name, passengerNo, department, email, address
from passenger;

create view passenger_contact as
select name, email from passenger;

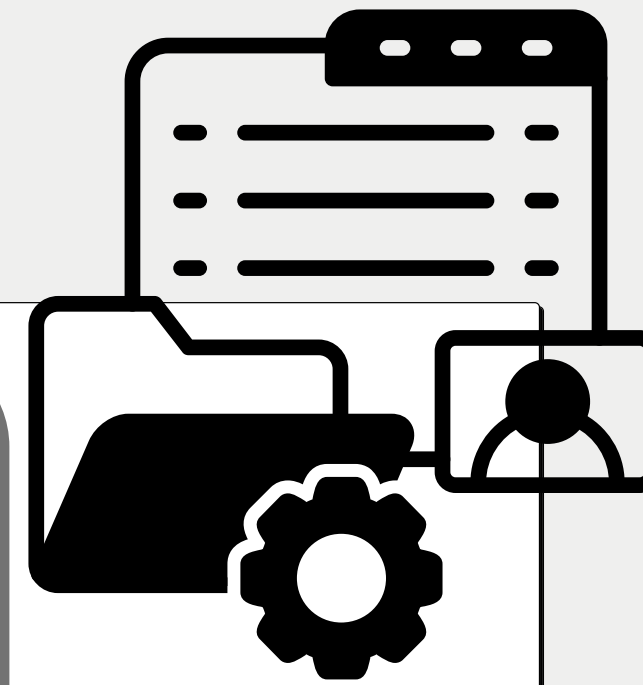
create view passenger_with_passport as
select passenger.name, passport.nationality, passport.expiry_date
from passenger
inner join passport on passenger.passengerId = passport.passengerId;

-- BOOKING VIEWS

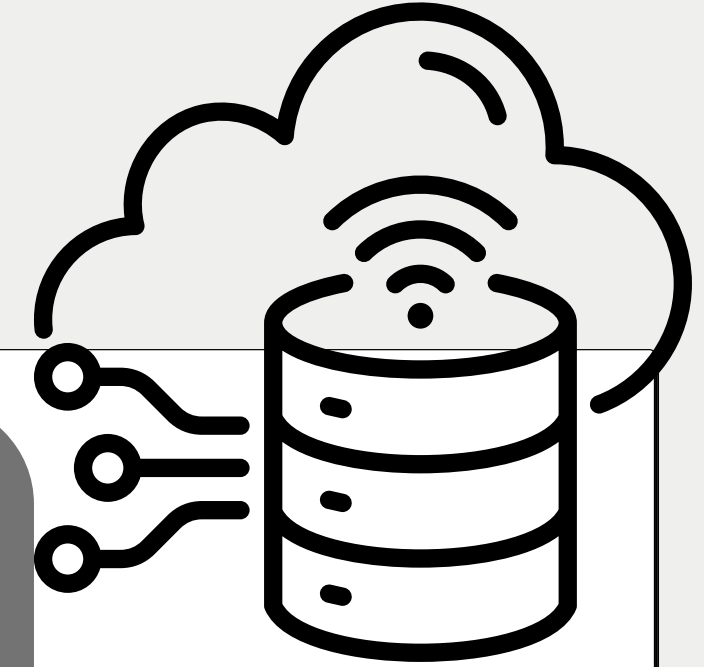
-- BOOKING VIEWS

create view booking_info as
select bookingId, paymentStatus, bookingDate, name as passenger_name, flightNo
from booking
inner join passenger on booking.passengerId = passenger.passengerId
inner join flight on booking.flightId = flight.flightId;

create view paid_bookings as
select * from booking where paymentStatus = 'paid';
```



Q TICKET & SECURITY CHECK VIEWS



```
-- TICKET VIEWS
```

```
create view ticket_info as  
select ticketNo, bookingId, issueDate, price, seatNo  
from ticket;
```

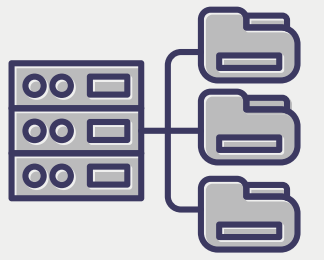
```
create view expensive_tickets as  
select * from ticket where price > 1500;
```

```
-- SECURITY CHECK VIEWS
```

```
create view security_summary as  
select passengerId, status, time  
from securityCheck;
```

```
create view hold_passengers as  
select passengerId, time  
from securityCheck  
where status = 'hold';
```

QPASSPORT & STAFF & BAGGAGE VIEWS



-- PASSPORT VIEWS

create view passport_info as
select passportId, nationality, issueDate, expiry_date, gender, passengerId
from passport;

create view expired_passports as
select * from passport where expiry_date < '2030-01-01';

-- BAGGAGE VIEWS

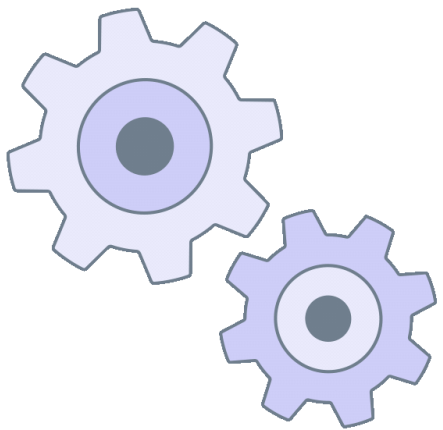
create view baggage_info as
select bag_no, passengerId, status, weight
from baggage;

create view lost_baggage as
select * from baggage where status = 'lost';

-- STAFF VIEWS

create view staff_info as
select staffId, role, name, department, email
from staff;

create view pilots as
select * from staff where role = 'Pilot';



Q WORKS_ON & ADDRESS TABLE VIEWS

```
-- works_on VIEWS
--
create view staff_flight_assign as
select staff.name, staff.role, flight.flightNo
from works_on
inner join staff on works_on.staffId = staff.staffId
inner join flight on works_on.flightId = flight.flightId;

-- ADDRESS TABLE VIEWS
--
create view passenger_addresses as
select passengerId, address from passenger_address;

create view staff_addresses as
select staffId, address from staff_address;
```





Lockdown Crew

Dec 25, 2025

Thank you!

دستم
سالمين